

Index

AalbmendoE, 276
 Abel, Niels Henrik, 410, 522
 Abel summable, 522
 Abel's formula for summation by parts, 392
 Abel's Lemma, 393
 Abel's test, 496
 Abel's Theorem, 522
 Absolute value, 11
 of a complex number, 534
 Absolutely convergent, 480, 556
 Absolutely summable, 480
 Acceleration, 161
Acta Eruditorum, 148
 Addition, 3
 associative law for, 9
 commutative law for, 9
 of complex numbers, 531
 geometric interpretation of, 535
 of vector-valued functions, 245
 of vectors, 75
 Addition formula
 for arcsin, 317
 for arctan, 317
 for cos, 314
 for sin, 313, 314
 for tan, 317
 Additive identity
 existence of, 9
 for vectors, 76
 Additive inverses
 existence of, 9
 Algebra, Fundamental Theorem of, 377, 538, 548, 567
 Algebraic functions, 363
 Algebraic number, 442
 Algebraist's real numbers, 598
 Almost lower bound, 142
 Almost upper bound, 142
 Analyst's real numbers, 598
 Angle, 303
 directed, 303
 Antidiagonal, 243
 Arabic numerals, multiplication of, 8
 Arc length, 278, 283
 Arccos, 311
 derivative of, 311
 Archimedes, 138, 141, 263
 Archimedean property
 for the rational numbers, 584
 for the real numbers, 138
 Archimedean spiral, 85, 249
 Arcsec, 320, 383
 Arcsin, 310
 addition formula for, 317
 derivative of, 311
 Taylor series for, 517
 Arctan, 311
 addition formula for, 317
 derivative of, 311
 Taylor polynomials for, 413, 420
 remainder term for, 426
 Area, 253, 258
 Arg cosh, 353
 Arg sinh, 353
 Arg tanh, 353
 Argument, 536
 Argument function, 542
 discontinuities of, 546
 Argument of the hyperbolic functions, 353
 Arithmetic mean, 33
 Arrow, 75, 76
 "x arrow $\sin(x^2)$ ", 45
 Associative law
 for addition, 9
 of vectors, 76
 for multiplication, 9
 Average velocity, 152
 Axis
 horizontal, 57
 imaginary, 534
 real, 533
 vertical, 57

 Bacon, Francis, vi
 Basic properties of numbers, 3
 "Bent graphs", 149
 Bernoulli, Jakob, 148, 574
 Bernoulli numbers, 572
 Bernoulli polynomials, 575
 Bernoulli's inequality, 32
 Big game hunting, mathematical theory of, 552
 Binary operation, 581

- Binomial coefficient, 27, 437
- Binomial series, 495, 518
- Binomial theorem, 28
- Bisection argument, 142, 552
- Bohr, Harold, 394
- Bolzano-Weierstrass Theorem, 458, 469, 552
- Bound
 - almost lower, 142
 - almost upper, 142
 - greatest lower, 134
 - least upper, 133, 584
 - lower, 134
 - upper, 133
- Boundedness criterion, 474
- Bounded above, 122, 133, 457, 584
- Bounded below, 134, 457
- Bourbaki, Nicholas, 148
- Cantor, Georg, 448
- Cardioid, 89, 250
- Cartesian coordinates, 84
- Cauchy
 - Condensation Theorem, 496
 - criterion, 473
 - form of the remainder, 437
 - Mean Value Theorem, 204
 - product, 493, 513
 - sequence, 459, 571
 - equivalence of, 599
- Cauchy-Hadamard formula, 569
- Cauchy-Schwarz inequality, 281
- Cavalieri, Bonaventura, 275
- Cesaro summable, 493
- Chain Rule, 174 ff.
 - proof of, 178
- Change, rate of, 152
- Characteristic (of a field), 586
- Circle, 65
 - " f circle g ", 44
 - unit, 66
- Circle of convergence, 560
- Classical notation
 - for derivatives, 154–156, 162, 167, 187, 241–242
 - for integrals, 265
- Cleio, 186
- Closed interval, 57
- Closed rectangle, 547
- Closure under addition, 9
- Closure under multiplication, 9
- Commutative law
 - for addition, 9
 - of vectors, 76
 - for multiplication, 9
- Comparison test, 474
- limit, 475
- Comparison Theorem, Sturm, 323
- Complete induction, 23
- Complete ordered field, 584, 603
- Completing the square, 17, 379
- Complex analysis, 565
- Complex function
 - continuous, 545, 546
 - differentiable, 550
 - graph of, 542
 - limit of, 542
 - nondifferentiable, 551
 - Taylor series for, 563
- Complex n th root, 536
- Complex numbers, 526, 531
 - absolute value of, 534
 - addition of, 531
 - geometric interpretation of, 535
 - geometric interpretation of, 533–534
 - imaginary part of, 531
 - infinite sequence of, 555
 - infinite series of, 555–557
 - logarithm of, 570
 - modulus of, 534
 - multiplication of, 531
 - geometric interpretation of, 535–536
 - real part of, 531
- Complex plane, 533
- Complex power series, 557
 - circle of convergence of, 560
 - radius of convergence of, 559
- Complex-valued functions, 541
- Composition of functions, 44
- Concave function, 220
- Conditionally convergent series, 481
- Cone, 80
 - generating line of, 80
 - surface area of, 404
- Conic sections, 80; *see also* Ellipse, Hyperbola, Parabola
- Conjugate, 534, 539

- Conjugate function, 541
- Constant function, 43
- Construction of the real numbers, 588 ff.
- Continued fraction, 462
- Continuous, uniformly, 144
- Continuous at a , 115, 545
- Continuous function, 115, 118, 546
 - nowhere differentiable, 159, 509
- Continuous on (a, b) , 118
- Continuous on $[a, b]$, 118
- Contraction, 466
- Contraction lemma, 466
- Converge
 - pointwise, 502
 - uniformly, 502, 506
- Convergent sequence, 453, 555
- Convergent series, 472, 556
 - absolutely, 480, 556
 - conditionally, 481
- Convex function, 219
 - strictly, 228
 - weakly, 228
- Convex subset of the plane, 229, 553
- Cooling, Newton's law of, 356
- Coordinate
 - first, 57
 - second, 57
- Coordinate system, 57
 - cartesian, 84
 - origin of, 57
- Coordinates
 - polar, 84
- "Corner", 60
- Cos, 303–304, 306, 321–322, 563
 - addition formula for, 314
 - derivative of, 172, 307
 - inverse of, *see* Arccos
 - Taylor polynomials for, 413
 - remainder term for, 424
- Cosh, 353
- Cosine, hyperbolic, 353
- Cot, 310
 - derivative of, 310
- Countable, 449
- Counting numbers, 21
- Critical point, 190
- Critical value, 190
- Csc, 310
 - derivative of, 310
- Cubic equation, general solution, 528–529
- Curve
 - parameterized, tangent line of, 246
 - parametric representation of, 244
 - reparameterization of, 247
- Cycloid, 250

- Darboux's Theorem, 214
- De Moivre's Theorem, 536
- Decimal expansion, 73, 492, 599
- Decreasing function, 195
- Decreasing sequence, 457
- Dedekind, Richard, 38
- Defined implicitly, 241
- Definite integral, 365
- DEFINITION**, 47
- Definition, recursive, 23
- Degree (of a polynomial), 42
- Degree measurement, 304–305
- Delicate ratio test, 493
- Delicate root test, 493
- Dense, 140
- Derivative, 149 ff., 151
 - classical notation for, 154–156, 162, 167, 187, 241–242
 - higher-order, 161
 - "infinite", 158
 - left-hand, 156
 - Leibnizian notation for, *see* Derivative, classical notation for
 - logarithmic, 351
 - "negative infinity", 158
 - of f , 151
 - of f at a , 151
 - of vector-valued function, 246
 - right-hand, 156
 - Schwarzian, 184
 - second, 161
 - Schwarz, 439
- Derivative of quotient, incantation for, 171
- Descartes, René, 84
- Determinant, 79
 - of vector-valued functions, 248
- Diagonal, 233

- Difference operator, 573
- Differentiable, 151, 550
- Differential equation, 292, 300, 321, 322–323, 356, 361, 438–440
 - initial conditions for, 440
- Differentiation, 168 ff.
 - implicit, 241
 - logarithmic, 351
- Differentiation operator, 573
- Dini's Theorem, 524
- Directed angle, 303
- Dirichlet's test, 495
- Disc method, 402
- Discontinuities of a nondecreasing function, 450
- Discontinuity, removable, 121
- Disraeli, Benjamin, 2
- Distance, 58, 534
 - shortest between two points, 278
- Distributive law, 9
- Diverge, 453, 556
- Division, 6
- Division by zero, 6
- Domain, 40, 41, 47, 601
- Dot product
 - of vectors, 78
 - of vector-valued functions, 246
- Double intersection, 165
- Double root, 185
- Durège, 38

- e , 343
 - irrationality of, 429
 - relation with π , 448, 564
 - transcendentality of, 444
 - value of, 344, 426
- Eccentricity of ellipse, 87
- Elementary function, 363
- Ellipse, 66, 82
 - axes of, 87
 - eccentricity of, 87
 - equation in polar coordinates, 86–87
 - focus point of, 66, 86
 - major axis of, 87
 - minor axis of, 87
- Ellipsoid of revolution, 405
- Empty collection, 23

- Entire function, 567
- Epsilon, 18
- Equal up to order n , 418
- Equality, order of, 418
- Equations, differential, *see* Differential equations
- Equivalent Cauchy sequences, 599
- Etymology lesson, 82
- Euler, Leonhard, 575
- Euler's number, 463
- Euler-Maclaurin Summation Formula, 576
- Even function, 51, 199
- Even number, 25
- Eventually inside, 555
- Exhaustion, method of, 141
- Exp, 343 ff., 563
 - classical approach to, 357
 - elementary definition of, 468
 - Taylor polynomials for, 413
 - remainder term for, 426
- Expansion, decimal, 73, 492, 599
- Extension of a function, 115–116

- Factorial, 23
- Factorials, table of, 432
- Factorization into primes, 31
- Fibonacci, 32
- Fibonacci Association, 32
- Fibonacci Quarterly, The*, 32
- Fibonacci sequence, 32, 521, 572
- Field, 581
 - characteristic of, 586
 - complete ordered, 584, 603
 - ordered, 583
- First coordinate, 57
- First Fundamental Theorem of Calculus, 285
- Fixed point of a function, 465
- Focus point, 66, 86
- Force, as vector, 76
- Four leaf clover, 88
- Fourier series, 318, 320, 323
- Fraction, continued, 462
- Function, 39, 47
 - absolute value, 541

Function (*continued*)

- argument, 542
 - discontinuities of, 546
 - complex valued, 541
 - composition of, 44
 - concave, 220
 - conjugate, 541
 - constant, 43
 - continuous, 115 ff.
 - convex, 219
 - critical point of, 190
 - critical value of, 190
 - decreasing, 195
 - derivative of, 149 ff.
 - differentiable, 151, 550
 - elementary, 363
 - entire, 567
 - even, 51, 199
 - exponential, 343–344
 - extension of, 115–116
 - fixed point of, 465
 - from A to B , 601
 - from real numbers to the plane, 244
 - graphs of, 57–65, 198, 542
 - hyperbolic, 353
 - identity, 43
 - imaginary part, 541
 - implicitly defined, 241
 - increasing, 195
 - integrable, 258
 - integral of, 258
 - inverse, 231 ff.
 - linear, 58
 - local maximum point of, 189
 - local minimum point of, 189
 - local strict maximum point of, 218
 - logarithm, 341, 346
 - maximum point of, 188
 - maximum value of, 188
 - minimum value of, 188
 - most general definition of, 601
 - negative part of, 51
 - nondecreasing, 243
 - nonincreasing, 243
 - nonnegative, 51
 - notation for, 40, 45
 - odd, 51, 199
 - one-one, 230
 - periodic, 71, 164, 298
 - polynomial, 42
 - positive part of, 51
 - power, 60
 - product of, 43
 - quotient of, 43
 - rational, 42
 - real part, 541
 - real-valued, 541
 - “reasonable”, 68, 118, 149, 180
 - regulated, 524
 - square root, 546–547
 - step, 278
 - strict maximum point of, 218
 - sum of, 43
 - trigonometric, 303 ff.
 - value at x , 40
 - vector-valued, 244
- Fundamental Theorem of Algebra, 377, 538, 549, 567
- Fundamental Theorem of Calculus
- First, 285
 - Second, 289
- Gabriel, 408
- Galileo Galilei, 148, 164
- Gamma function, 394, 444
- Generating line, of a cone, 80
- Geometric mean, 33
- Geometric series, 473
- Global property, 123
- Goes to, “ x goes to $\sin(x^2)$ ”, 45
- Graph of polynomial function, 197–198
- Graph sketching, 196–201
- Graphs, 57–65, 85 ff., 90–91, 197–198, 542
- Gravitation, 330
- Greatest lower bound, 134
- Grin and bear it, 385–386
- Gronwall’s inequality, 356
- Grow
- at the same rate as, 361
 - faster than, 361

- Hadamard, *see* Cauchy-Hadamard formula
- Half-life (of radioactive substance), 356
- Hermite, Charles, 443
- High-school student's real numbers, 599
- Higher-order derivatives, 161
- Hilbert, David, 443
- Horizontal axis, 57
- Hyperbola, 67, 82
 - equation in polar coordinates, 88
- Hyperbolic cosine, 353
- Hyperbolic functions, 353
- Hyperbolic sine, 353
- Hyperbolic spiral, 316
- Hyperbolic tangent, 353

- Identity
 - additive, 9
 - multiplicative, 9
- Identity function, 43
- Identity operator, 574
- Imaginary axis, 534
- Imaginary part function, 541
- Imaginary part of a complex number, 531
- Implicit differentiation, 241
- Implicitly defined, 241
- Improper integral, 301–302, 397–399
- Incantation for derivative of quotient, 171
- Increasing at a , 217
- Increasing function, 195
- Increasing sequence, 457
- Indefinite integral, 365
- Indefinite integrals, short table of, 365–366
- Induction, mathematical, 21
 - complete, 23
- Inductive set of real numbers, 34
- Inequalities, 9
 - in an ordered field, 584
- Inequality
 - Bernoulli's, 32
 - Cauchy-Schwarz, 281
 - geometric-arithmetic mean, 33
 - Gronwall's, 356
 - Jensen's, 228
 - Liouville's, 448
 - Schwarz, 17, 33, 281
 - triangle, 71
 - Young's, 276
- Infimum, 134
- "Infinite" derivative, 158
- Infinite intervals, 57
- Infinite product, 329, 395, 497
- Infinite sequence, 452, 555
- Infinite series, 471
 - multiplication of, 486
- Infinite sum, 430, 471
- Infinite trumpet, 408
- Infinitely many primes, 32
- "Infinitely small", 155, 264
- Infinity, 57
 - minus, 57
- Inflection point, 225
- Initial conditions for differential equations, 440
- Instantaneous speed, 152
- Instantaneous velocity, 152
- Integer, 25
- Integrable, 258
- Integral, 258
 - classical notation for, 264–265
 - definite, 365
 - improper, 301–302, 397–399
 - indefinite, 365
 - short table of, 365–366
 - Leibnizian notation for, *see* Integral, classical notation for
 - lower, 295
 - Mean Value Theorem for, 277
 - Second Mean Value Theorem for, 391
 - upper, 295
- Integral form of the remainder, 423
- Integral sign, 258
- Integral test, 478
- Integration
 - by parts, 366 ff.
 - by substitution, 369 ff.
 - limits of, 258
 - of rational functions, 377 ff.
- Interest (finance), 355

- Intermediate Value Theorem, 124, 131, 135, 299
- Interpolation, Lagrange, 49
- Intersection of sets, 43
- Interval, 56
 - closed, 57
 - infinite, 57
 - open, 56; *see also* Nested Intervals Theorem
- Inverse
 - additive, 9
 - multiplicative, 9
- Inverse of a function, 231 ff.
- Inverse square law, 330
- Inverses of trigonometric functions, *see* Trigonometric functions
- Irrational numbers, 25
- Isomorphic fields, 602
- Isomorphism, 602

- Jensen's inequality, 228
- Johnson, Samuel, 607
- Jump, 60

- Kepler, Johannes, 330
- Kepler's laws of planetary motion, 330

- Lagrange form of the remainder, 423, 436
- Lagrange interpolation formula, 49
- Large negative, 64
- Least upper bound, 133 ff., 584
- Least upper bound property, 135
- Lebesgue, *see* Riemann-Lebesgue Lemma
- Left-hand derivative, 156
- Leibniz, Gottfried Wilhelm, 155, 264
- Leibniz's formula, 184
- Leibniz's Theorem, 481
- Leibnizian notation for derivatives, 154–156, 167, 187, 241
 - for higher order derivatives, 162
- Lemma, 102
- Lemniscate, 89
- Length, 278, 283
- L'Hôpital, Marquis de, 148
- L'Hôpital's Rule, 204, 213–214
- Limit, 90 ff., 98, 542
 - at infinity, 106
 - "does not exist", 101
 - from above, 106
 - from below, 106
 - of a sequence, 453
 - of vector-valued function, 246, 252
 - uniqueness of, 100
- Limit comparison test, 475
- Limit of integration, 258
- Limit point, 469, 552
- Limit superior, 143, 467
- Lindemann, Ferdinand von, 447
- Line, real, 56
- Line, tangent, *see* Tangent line
- Linear functions, 58
- Liouville, Joseph, 448
- Liouville's inequality, 448
- Liouville's Theorem, 567
- Lipschitz of order α , 210
- Local maximum point of function, 189
 - higher-order derivative test for, 417
 - second derivative test for, 201
- Local minimum point of function, 189; *see also* Local maximum point
- Local property, 109, 123, 166
- Local strict maximum point, 218
- Log, 341, 346
 - Taylor polynomials for, 413
 - remainder term for, 427
- Logarithm
 - classical approach to, 357
 - Napierian, 358
 - of a complex number, 570
 - to the base 10, 339
- Logarithmic derivative, 351
- Lower bound, 134
 - almost, 142
 - greatest, 134
- Lower integral, 295

- Lower limit of integration, 258
- Lower sum, 254
- Lowest terms, 73

- Maclaurin, *see* Euler-Maclaurin summation formula
- Major axis of ellipse, 87
- Mass, rate of change of, 152
- Mathematical induction, 21
- Maximum of two numbers, 16
- Maximum point of a function, 188
 - local, 189; *see also* Local maximum point
 - local strict, 218
 - strict, 218
- Maximum value of function, 188
- Mean
 - arithmetic, 33
 - geometric, 33
- Mean Value Theorem, 193, 194
 - Cauchy, 204
 - for integrals, 277
 - Second, 391
- Method of exhaustion, 141
- Minimum of function, 188
- Minimum of two numbers, 16
- Minimum point of a function, local, 189; *see also* Local minimum point
- Minor axis of ellipse, 87
- Minus infinity, 57
- Mirifici logarithmonum canonicis description*, 358
- Modulus of a complex number, 534
- Møllerup, Johannes, 394
- Multiplication, 5
 - associative law for, 9
 - closure under, 9
 - commutative law for, 9
 - of arabic numerals, 8
 - of complex numbers, 531
 - geometric interpretation, 535–536
 - of function and vector-valued function, 245
 - of infinite series, 486
 - of number and vector, 77
 - of vectors, 77
- Multiplicative identity, existence of, 9
- Multiplicative inverses, existence of, 9
- Multiplicity (of a root), 130, 185–186

- Napier, John, 358
- Napierian logarithm, 358
- Natural numbers, 21, 34
- Negative, large, 64
- “Negative infinity”, derivative, 158
- Negative number, 9
- Negative numbers, product of two, 7
- Negative part of a function, 51
- Nested Interval Theorem, 142
- Newton, Isaac, 155, 276, 330
- Newton’s law of cooling, 356
- Newton’s laws of motion, 161
- Newton’s method, 464
- Nondecreasing function, 243
- Nondecreasing sequence, 457
- Nondifferentiable complex functions, 551
- Nonincreasing function, 243
- Nonincreasing sequence, 457
- Nonnegative function, 51
- Nonnegative sequence, 474
- Norm, 78, 252
- Notational nonsense, 573
- Nowhere differentiable continuous function, 509
- n th root, 71, 536
 - existence of, 125, 536, 553
 - primitive, 540
- Null set, 23
- Number
 - algebraic, 442
 - complex, 526, 531
 - counting, 21
 - even, 25
 - imaginary, 526
 - irrational, 25
 - natural, 21, 34
 - odd, 25
 - prime, 31
 - rational, 25
 - real, 25, 534, 589
 - transcendental, 442
- Numbers, basic properties of, 3

- Odd function, 51, 199
- Odd number, 25
- One-one function, 230
- Open interval, 56
- “Or”, 6
- Order of equality, 418
- Ordered field, 583
 - complete, 584
- Ordered pair, 47 (footnote), 54
- Origin (of a coordinate system), 57

- Pair, 46
 - ordered, 47 (footnote), 54
- Parabola, 60, 82
 - area under, 263
 - equation in polar coordinates, 88
- Parallelogram, 76
- Parameterized curve, tangent line of, 246
- Parametric representation of a curve, 244
- Partial fraction decomposition, 378
- Partial sums, 471
- Partition, 254
- Parts
 - Abel’s formula for summation by, 392
 - integration by, 366 ff.
- Pascal’s triangle, 27
- “Peak”, 61
- Peak point, 458
- Period of a function, 71, 164, 298
- Periodic function, 71, 164, 298
- Perpendicularity of lines, 70
- Petard, H., 552
- Pig, yellow, v, 375
- Pigheaded, 186
- Plane, 58
 - complex, 533
- Planetary motion, Kepler’s laws of, 330
- Point, 56
- Point of contact, 220
- Point-slope form of equation of a line, 59, 70
- Polar coordinates, 84 ff.
- Polynomial function, 42
 - graph of, 61, 197 ff.
 - multiplicity of roots, 130, 185–186
- Polynomials, Bernoulli, 575
- Pope, Alexander, 330
- Position, rate of change of, 152
- Positive element of $\mathbf{R}$, 593
- Positive elements of an ordered field, 583
- Positive number, 9
- Positive part of a function, 51
- Power functions, 60
- Power series, 510
 - complex, 557
 - centered at a , 510, 564
- Powers of 2, table of, 432
- Prime number, 31
 - characteristic of a field, 586
 - infinitely many of, 32
 - unique factorization into, 31
- Primitive, 363
- Primitive n th root, 540
- Principia*, 276
- Product, 5
 - Cauchy, 493, 513
 - infinite, 329, 395, 497
 - of function and vector-valued function, 245
 - of functions, 43
 - of number and vector, 77
 - of two negative numbers, 7
 - of vectors, 77
- Pyramid
 - surface area of, 403
 - volume of, 407
- Pythagorean theorem, 25, 58
- π , 305
 - Archimedes’ approximation of, 141
 - irrationality of, 326
 - relation to e , 448, 564
 - transcendentality of, 447
 - value of, 433
 - Viète’s product for $2/\pi$, 329
 - Wallis’ product for $\pi/2$, 395

- Quaternions, 587
- Quotient of functions, 43

- Rabbits
 - growth of population, 32
- Radian measure, 63, 304–305
- Radioactive decay, 355–356
- Radius of convergence of complex power series, 559
- Rate of change of mass, 152
- Rate of change of position, 152
- Ratio test, 476
 - delicate, 493
- Rational functions, 42
 - integration of, 377 ff.
- Rational numbers, 25
- Real axis, 533
- Real line, 56
- Real number (formal definition), 589
- Real numbers, 25
 - algebraist's, 598
 - analyst's, 598
 - Archimedian property of, 138
 - construction of, 588 ff.
 - high-school student's, 599
 - inductive set of, 34
- Real part function, 541
- Real part of a complex number, 531
- Real-valued function, 541
- Rearrangement of a sequence, 483
- “Reasonable” function, 68, 118, 149, 180
- Rectangle, closed, 547
- Recursive definition, 23
- Reduction formulas, 377
- Regulated function, 524
- Remainder term for Taylor polynomials, 421, 423, 437
- Removable discontinuity, 121
- Reparameterization, 247
- Revolution
 - ellipsoid of, 405
 - solid of, 402
- Riemann sum, 282
- Riemann-Lebesgue Lemma, 320, 391
- Right-hand derivative, 156
- Rising Sun Lemma, 143
- Rolle, Michel, 186
- Rolle's Theorem, 193
- Root
 - multiplicity of, 130, 185–186
- Root of a polynomial function, 50
 - double, 185; *see also* *n*th roots
- Root test, 493
 - delicate, 493
- Same sign, 12
- Scalar, 78
- Scalar product of vectors, 78
- Schwarz, H. A., 217
- Schwarz inequality, 17, 33, 281
- Schwarz second derivative, 438
- Schwarzian derivative, 184
- Sec, 310
 - derivative of, 310
 - inverse of, *see* Arcsec
- Secant line, 150
- Second coordinate, 57
- Second derivative, 161
 - Schwarz, 438
- Second derivative test for maxima and minima, 201
- Second Fundamental Theorem of Calculus, 289
- Second Mean Value Theorem for Integrals, 391
- Sequence
 - absolutely summable, 480
 - Cauchy, 459
 - complex, 571
 - equivalence of, 599
 - complex numbers, 555
 - convergent, 453
 - pointwise, 502
 - uniformly, 502
 - decreasing, 457
 - divergent, 453, 556
 - Fibonacci, 32, 521, 572
 - increasing, 457
 - infinite, 452, 555
 - limit of, 453
 - nondecreasing, 457
 - nonincreasing, 457
 - nonnegative, 474
 - rearrangement of, 483
 - summable, 472
- Series
 - absolutely convergent, 480
 - conditionally convergent, 481

- Series (*continued*)
 - convergent, 472, 556
 - Fourier, 318, 320, 323
 - geometric, 473
 - power, 510, 557
 - Taylor, 511
- Set, 22
 - empty, 23
- Sets
 - intersection of, 43
 - notation for, 43–44
- Shadow point, 143
- Shell method, 403
- Sigma, 24
- Sign, 12
- Simpson's rule, 401
- Sin, 43, 303–304, 306, 321–322, 563
 - addition formula for, 313, 314
 - derivative of, 172, 307
 - inverse of, *see* Arcsin
 - Taylor polynomials for, 412
 - remainder term for, 424
- Sine, hyperbolic, 353
- Sine function, 43
- Sinh, 353
- Sketching graphs, 196–201
- Skew field, 587
- Slope of a straight line, 58
- Solid of revolution, 402
- Speed, instantaneous, 152
- Spiral
 - Archimedian, 85, 249
 - hyperbolic, 316
- Square root, 12, 527
 - existence of, 124
- Square root function, 546–547
- Square root in a field, 586
- Squaring the circle, 447
- Step function, 278
- Stirling's Formula, 578
- Straight line
 - analytic definition, 58
 - shortest distance between two points, 278
 - slope of, 58
- Strict maximum point, 218
- Strictly convex, 228
- Sturm Comparison Theorem, 323
- Subsequence, 458
- Substitution
 - integration by, 369 ff.
 - world's sneakiest, 386
- Substitution formula, 369
- Subtraction, 5
- Sum
 - finite, 3–4
 - infinite, 430, 471
 - lower, 254
 - of an infinite sequence, 472
 - of an infinite sequence of complex numbers, 555
 - of functions, 43
 - of vector-valued functions, 245
 - of vectors, 75
 - partial, 471
 - sigma notation for, 24
 - upper, 254
- Sum of squares, 552
- Summable, 472, 556
 - Abel, 522
 - absolutely, 480
 - Cesaro, 493
 - uniformly, 506
- Summation by parts, Abel's formula for, 392
- Supremum, 134
- Surface area
 - of cone, 404
 - of pyramid, 403
 - of solid of revolution, 404
- Swift, Jonathan, 580
- Symmetry in graphs, 199
- Tan, 310
 - derivative of, 310
 - inverse of, *see* Arctan
 - Taylor series for, 573
- Tangent, hyperbolic, 353
- Tangent line, 149, 151
 - of parameterized curve, 246
 - point of contact of, 220
 - "Tangent line", vertical, 158
- Tanh, 353
- Taylor polynomial, 412 ff.
 - remainder term of, 421, 423, 427; *see also specific functions*

- Taylor series, 511
- Taylor's Theorem, 424
- Torus, 405
- Transcendental number, 442
- Trapezoid rule, 400
- Triangle inequality, 71
- Trichotomy law, 9
- Trigonometric functions, 303 ff., *see also*
 - cos, cot, csc, sec, sin, tan
 - integration of, 376–377
 - inverses of, 310 ff., *see also* arccos, arcsec, arcsin, arctan
- Trumpet
 - infinite, 408
- Truncation of a polynomial, 434
- Two-time differentiable, 161

- Uniform limit, 502
- Uniformly continuous function, 144
- Uniformly convergent sequence, 502
- Uniformly convergent series, 506
- Uniformly distributed sequence, 469
- Uniformly summable, 506
- Uniqueness
 - of factorization into primes, 31
 - of limits, 100
- Unit circle, 66
- Upper bound, 133, 584
 - almost, 142
 - least, 133
- Upper integral, 295
- Upper limit of integration, 258
- Upper sum, 254

- “Valley”, 61
- Value
 - absolute, *see* Absolute value
- Value of f at x , 40

- Vanishing condition, 473
- Vector-valued functions, 244
- Vector-valued functions
 - determinant of, 246
 - derivative of, 246
 - dot product of, 246
 - limit of, 246, 252
 - multiplication of function by, 245
 - sum of, 245
- Vectors, 75
 - addition of, 75
 - as forces, 76
 - dot product of, 78
 - multiplication by numbers, 77
 - multiplication of, 77
 - scalar product of, 78
- Velocity
 - average, 152
 - instantaneous, 152
- Vertical axis, 57
- Viète, François, 329
- Volume, 402–403
 - of solid of revolution, 402

- Wallis' product, 395
- Weakly convex, 228
- Weierstrass, *see* Bolzano-Weierstrass Theorem
- Weierstrass M -test, 507
- Well-ordering principle, 23
- Wright, Edward, 388

- Young's inequality, 276

- Zahl, 25
- Zero, division by, 6